1a) Estable $\Leftrightarrow |z| = 1 \in$ ROC

$\Rightarrow \beta \in [0,1)$ y $\gamma \in (1, +\infty)$

Será causal si $\gamma \to \infty$

b) Linealidad: $X[n] = z[n] + b[n]$

$w[n] = a[n] \cdot \alpha^n + b[n] \cdot \alpha^n$

$\Rightarrow$ Si h es LTI, listo.

Invariante en el tiempo:

$$y[n] = w[n] * h[n] = \sum_{k \in \mathbb{Z}} h[k]\, w[n-k] = \sum_{k \in \mathbb{Z}} h[k] \alpha^{n-k} x[n-k]$$

$$y[n] = \alpha^{-n} z[n]$$

$$= \sum_{k \in \mathbb{Z}} h[k] \alpha^{-k} x[n-k]$$

} salida del sistema $y[n]$

$$y[n-n_0] = \sum_{k \in \mathbb{Z}} h[k] \alpha^{-k} x[n-n_0-k]$$

Si entra $x[n-n_0] \Rightarrow$ la salida es $y[n-n_0]$

$\Rightarrow$ Invariante en el tiempo.

11-8-14

1a) BIBO estable: Salida acotada ∀ entrada acotada

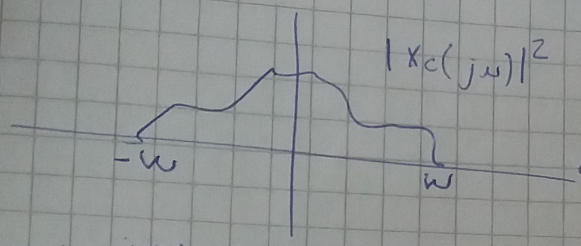
$$y[n] = \sum_{k\in\mathbb{Z}} h[k]\,\alpha^{k}\,x[n-k]$$

- Si $\beta = 0$ y $\gamma \to \infty$ significa longitud de h acotada → pero $\forall x$
- Si $\beta > 0$ y $\gamma \to \infty \Rightarrow h$ se extiende infinitamente hacia la derecha

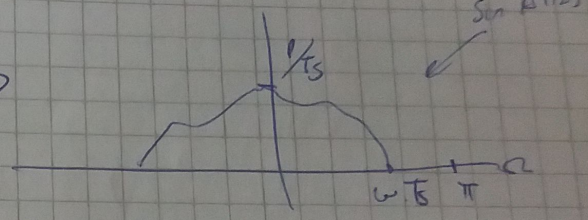
se requiere que $|\alpha| \geq 1$

- Si $\beta = 0$ y $\gamma < \infty \Rightarrow h$ es finita $\Rightarrow |\alpha| < 1$

2) $$E_d = \sum_{k\in\mathbb{Z}} |x[n]|^2 = \frac{1}{2\pi}\int_{2\pi} |X(e^{j\Omega})|^2\, d\Omega \leftarrow \text{Parseval}$$



$\Rightarrow$



$$X(e^{j\Omega}) = X_c\left(j\frac{\Omega}{T_s}\right)\frac{1}{T_s} \rightarrow \Omega \in [-\pi;\pi]$$

$$\Rightarrow E_d = \frac{1}{2\pi}\int_{-\omega T_s}^{\omega T_s} \frac{1}{T_s^2}\,|X_c(j\tfrac{\Omega}{T_s})|^2\, d\Omega$$

$$\begin{cases} \omega = \frac{\Omega}{T_s} \\ d\omega = \frac{d\Omega}{T_s} \end{cases} \Rightarrow \frac{1}{2\pi T_s}\int |X_c(j\omega)|^2 d\omega \Rightarrow E_d = \frac{E_c}{T_s}$$

3) $$x[n]\,w[n] = \frac{x[n]}{2} + \frac{x[n]}{2}\cos\left(\frac{2\pi}{N}\left(n-\frac{N}{2}\right)\right)$$

$$= \frac{x[n]}{2} + \frac{x[n]}{4}\left\{\exp\left(j\frac{2\pi}{N}\left(n-\frac{N}{2}\right)\right) + \frac{x[n]}{4}\exp\left(-j\frac{2\pi}{N}\left(n-\frac{N}{2}\right)\right)\right.$$

$$\Rightarrow \underset{N}{DFT}\left\{\tfrac{1}{2}x[n]\,w[n]\right\} = \frac{X[k]}{2} + \exp(-j\pi)\frac{X}{4}\left[(k+1)_N\right] + \exp(j\pi)\frac{X}{4}\left[(k+1)_N\right]$$

$$= \frac{X[k]}{2} - \frac{1}{4}\left[X\left[(k+1)_N\right] + X\left[(k-1)_N\right]\right]$$

Falta 2